

WEN-HORNG SHEU

Contact: (530) 979-6045 ♦ wsheu@ucdavis.edu

Personal Website: <https://whsheu.github.io/>

LinkedIn: <https://www.linkedin.com/in/whsheu/>

Professional Summary: Ph.D. candidate in Computer Science specializing in algorithms for large-scale and memory-constrained computation. Over five years of research experience and a strong background in competitive programming.

EDUCATION

Ph.D. in Computer Science, University of California, Davis; GPA: 4.0/4.0 2023-Present

- Expected Graduation Date: June 2028.
- Research Area: Distributed Algorithms, Streaming Algorithms.

M.S. in Computer Science, National Tsing Hua University; GPA: 3.9/4.0 2019-2021

- Selected Coursework: Approximation Algorithms, Computational Geometry.

B.S. in Computer Science, National Tsing Hua University; GPA: 3.85/4.0 2015-2019

- Selected Coursework: Advanced Data Structures, Randomized Algorithms, Parallel Algorithm Design.

SELECTED PUBLICATIONS

Fixed-Threshold Peeling in Sublinear MPC: Round-Approximation Tradeoffs and Applications, DISC '26

- Designed distributed algorithms for the edge orientation, k -core, and densest subgraph problems.
- Improved the round complexity from $\log^{1/2} n$ to $\log^{1/4} n$ via a pruned graph exploration technique.

Faster Semi-streaming Matchings via Alternating Trees, ICALP '25

- Designed an algorithm for computing an approximate matching in memory-constrained settings.
- Improved a STOC '22 result, reducing pass complexity from $O(\epsilon^{-19})$ to $O(\epsilon^{-6})$ via a new data structure.

Faster MPC Algorithms for Approximate Allocation and Matching in Uniformly Sparse Graphs, SPAA '25

- Developed a new distributed algorithm with applications in online advertising and load balancing.
- Reduced round complexity from $O(\log n)$ to $O(\sqrt{\log n})$ using random sampling, extending an ICML '18 result.

SELECTED HONORS

- **Shadow Program Committee Member** for ACM SPAA 2026
 - Invited to serve on the shadow program committee for a leading conference in parallel algorithms.
 - Reviewed 6 submissions; participated in committee discussions and contributed to paper evaluations.
- **Gold Award** in the 2019 ICPC Asia Pacific Taipei-Hsinchu Regional Contest
 - Competed in 6+ international programming contests, including multiple ICPC regional contests.
 - Solved 1,500+ algorithmic programming problems across graph algorithms and data structures.
- **Grandmaster on Codeforces**
 - Placed in the top 1% among 10,000+ global contestants in four Codeforces programming contests.
 - Ranked as a Grandmaster, within the top 1% globally on Codeforces as of February 2021.

SKILLS

Programming Languages

C++, Python

Tools

Git, L^AT_EX

Languages

English (fluent, TOEFL 105/120), Chinese (native)

Coursera Certificate

AI Agents and Agentic AI with Python & Generative AI